$p_d = 0$ (and thus there is extreme ideological segregation on the network); and (ii) an island model with *maximal connectivity*, where $p_s = p_d$ (and there is minimal homophily and no segregation by ideology). Our second main result is:

Theorem 3. *There exists $\bar{\varepsilon} > 0$ such that if $\varepsilon < \bar{\varepsilon}$, the platform's profit-maximizing sharing network is determined by a reliability threshold $r_P \in (0, 1)$ such that:*

(i) If $r < r_P$, the platform's profit-maximizing sharing network has maximal homophily.

(ii) If $r > r_P$, the platform's profit-maximizing sharing network has maximal connectivity.

Part (i) of the theorem shows that when articles are mostly unreliable — and likely to contain misinformation — the platform creates an extreme filter by designing its algorithms to achieve a sharing network with the greatest homophily. In contrast, part (ii) demonstrates that when articles have higher reliability, the platform refrains from introducing algorithmic homophily. This result highlights an important channel by which misinformation spreads: it is precisely when articles are likely to contain misinformation that the platform seeks to maximize engagement by creating (endogenous) echo chambers, or filter bubbles, where these articles spread virally within like-minded communities. Put differently, with low reliability content, neither the platform nor the users are disciplined about sharing misinformation, and so these news items spread virtually uninhibited.

It is worth noting that this theorem builds on but also significantly strengthens Theorem 2. In Theorem 2, the effects of homophily are non-monotone and are ambiguous when an article is neither very low reliability nor very high reliability ($r \in (\underline{r}, \bar{r})$). In contrast, Theorem 3 gives a sharp characterization of the platform's algorithm: when $r > r_P$, the platform goes for maximal connectivity, and when $r < r_P$, it chooses maximal homophily, and in this case, misinformation spreads virally precisely because of the echo chambers that the platform has manufactured. In both cases, the island structure of the network considered in Section 4 arises endogenously as the profit-maximizing sharing network for the platform.

In addition, Theorem 3 shows that when a platform can shape the network topology through its recommendation algorithm, the echo chamber effect that arises from Theorem 2(a) is exactly what fuels misinformation (whereas in Theorem 2(b), echo chambers were harmless). This corroborates and extends much of the previous literature mentioned earlier on the interaction between recommender systems, echo chambers, and misinformation.

Finally, it is notable that the platform designs its algorithms to generate filter bubbles within ideologically-congruent groups precisely when the relevant content is low-reliability and likely to contain misinformation. This observation generalizes to cases where the platform has less precise microtargeting technology, albeit in a less sharp way than the result presented in Theorem 3.[18]

[18] In this case, the platform still induces echo chamber-like environments to generate viral spread of low-reliability content, even if its choice does not take the form of an island network. For instance, if there are only three communities (with broad ideology spectra), a misinformation right-wing article can still spread virally via the usage of filter bubbles. However, the platform may now prefer a sharing network that lies outside the class of island networks considered in Section 4. Specifically, user engagement may be maximized if the left-wing community is completely disconnected from all other communities, but the moderate (middle) community has sparse connections to the right-wing community. This facilitates a strong echo chamber within the right-wing community but also allows the article to spread to the more moderate community (while receiving no discipline from the left-wing community).